

HighPhytoSparseNet: A Lightweight Multi-Head Object Detection Model for Edge-Based Agricultural Applications

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Abstract. Object detection in agricultural environments presents unique challenges, including scale variation, class imbalance, and limited annotated data. Although modern deep learning-based detectors achieve high performance on large-scale benchmarks, their computational complexity often restricts deployment on edge devices. In this work, we propose HighPhytoSparseNet, a lightweight anchor-based single-stage object detection framework designed for real-time botanical applications. The proposed architecture employs a MobileNetV3-Small backbone for efficient feature extraction, followed by a bidirectional feature pyramid network to enhance multi-scale representation. A three-head detection module is introduced to specialize in small, medium, and large object detection, with a head-weighted loss strategy to improve sensitivity toward fine plant structures. The model contains approximately 3.8 million parameters and requires only 0.5 GFLOPs, making it suitable for edge deployment. Experimental results demonstrate that HighPhytoSparseNet achieves 65.1% *mAP*@0.5 while maintaining low computational overhead, highlighting its effectiveness for real-time agricultural monitoring systems.

Keywords: Object detection · lightweight neural networks · edge computing · agricultural monitoring · multi-scale detection · MobileNetV3.

1 Introduction

Recent advancements in artificial intelligence have significantly accelerated the development of object detection methods within computer vision. In particular, agricultural monitoring has emerged as a critical application domain, where automated visual analysis enables crop health assessment, yield estimation, and disease detection. Despite substantial progress in detection accuracy, deploying such models efficiently on resource-constrained edge devices remains a significant challenge due to computational and energy limitations.

Saqib et al. [1] provide a comprehensive review of deep learning-based object detection techniques for agricultural applications, highlighting the evolution of

two-stage detectors derived from R-CNN and their region proposal mechanisms. In contrast, single-stage detectors such as YOLO (You Only Look Once) [2] perform localization and classification in a unified forward pass, enabling faster inference suitable for real-time applications. Chetan et al. [3] further explore YOLO variants for agricultural object detection, demonstrating their effectiveness for edge-oriented deployment. More recent approaches incorporate transformer-based enhancements to improve detection robustness and feature representation. Xue et al. [4] propose MTYOLOX, a multi-transformer-enabled architecture for tree-level apple inflorescence detection, while Helizani et al. [5] employ YOLOv3-tiny for coffee fruit detection during harvest, emphasizing lightweight design considerations.

In practical agricultural deployments, sensing units are often mounted on battery-powered robots, unmanned ground vehicles, or low-cost fixed installations with limited compute, memory, and network connectivity. Reliance on cloud inference is frequently infeasible due to unstable rural connectivity and strict latency requirements for real-time actuation (e.g., targeted spraying or harvesting). These constraints make lightweight, on-device detection models particularly attractive, as they reduce hardware cost and energy consumption while enabling continuous monitoring at the field edge. Although modern object detection frameworks are well-developed and adaptable to various agricultural scenarios, a fundamental trade-off persists between detection accuracy and computational efficiency. Many high-performance models incur substantial parameter counts and inference costs, limiting their suitability for real-time deployment on edge hardware. Consequently, achieving an optimal balance between accuracy, model complexity, and inference speed remains an open research challenge.

In this work, we introduce **HighPhytoSparseNet**, a lightweight object detection framework designed for efficient agricultural edge deployment, specifically targeting real-time detection of tomato stems and fruits under field-like conditions. The term “Sparse” in HighPhytoSparseNet reflects the model’s aggressively constrained parameter and compute budget, rather than the use of explicit sparse convolutional operators. In other words, sparsity is realized at the architectural level (3.8M parameters, 0.5 GFLOPs) to match the tight memory and power envelopes of edge devices deployed in greenhouses and fields. A custom agricultural dataset was constructed comprising over 2,500 training images along with 100 validation and 100 test samples, including annotated instances of tomato fruits and stems captured under varying environmental conditions. The model incorporates a MobileNetV3-Small backbone, a bidirectional FPN+PANet neck, three scale-specific detection heads, and a domain-adapted loss function combining CIoU, Focal Loss, and class-weighted supervision to address class imbalance and small-object detection challenges.

Beyond generic lightweight backbones, recent work from our group has explored hardware-aware and quantization-friendly designs for efficient vision models on embedded platforms, including SIMD-based acceleration for quantized DNNs and embedded vision transformers with simplified attention and optimized buffers [6, 7]. These studies emphasize that accuracy, memory access, and

hardware utilization must be considered jointly when targeting real-time inference on low-power devices.

2 Related Work

The related work consists of Two-Stage Object Detectors, Single-Stage Object Detectors, Lightweight and edge-oriented architectures. Two-stage detectors decompose detection into region proposal generation followed by classification and bounding box regression. R-CNN [8] applied deep CNNs to candidate regions from selective search, though repeated convolutional processing introduced redundancy. SPP-Net and Fast R-CNN [9] improved feature sharing and unified training. Faster R-CNN’s Region Proposal Network enabled end-to-end training, while R-FCN introduced position-sensitive score maps. Despite these optimizations, sequential proposal refinement limits real-time edge suitability [10].

Single-stage detectors eliminate explicit region proposals, performing classification and localization in one forward pass. YOLO [11] formulates detection as a regression problem over grid predictions. While fast, early versions struggled with small objects; later variants including YOLOv5 [12] improved accuracy and efficiency. SSD [13] introduced multi-scale feature maps with anchor boxes, improving small-object detection. RetinaNet [14] addressed class imbalance via focal loss, achieving two-stage accuracy at single-stage speed. These streamlined architectures are well-suited for real-time edge applications.

Efficient backbone design is central to edge deployment. MobileNet [15] introduced depthwise separable convolutions with width/resolution multipliers for flexible accuracy-size trade-offs. MobileNetV3 further optimized via neural architecture search, squeeze-and-excitation modules, and hard-swish activations. ShuffleNet [16] employed group convolutions and channel shuffling to reduce parameters while preserving representational capacity. At the framework level, EfficientDet [17] introduced compound scaling with BiFPN for state-of-the-art accuracy at reduced parameter counts. GAN-based approaches [18] can improve robustness under limited data or domain shift, but add training complexity unsuitable for lightweight edge deployment. Most existing models excel on large benchmark datasets but degrade on rare classes, limited data, or small objects, challenges that are pronounced in agricultural and industrial scenarios. To improve robustness in such domains, Ge et al. introduced YOLO-MSD, a multi-scale industrial surface defect detector tailored to fine-grained texture patterns [19], illustrating the benefit of domain-specific multi-scale feature fusion.

To address these challenges in agricultural monitoring, we propose **HighPhytoSparseNet**, a lightweight CNN-based detection framework with a multi-head detection and classification strategy for plant-based applications.

3 Proposed HighPhytoSparseNet Architecture

3.1 Overview and the detailed functions

HighPhytoSparseNet is a lightweight multi-scale object detection framework designed for real-time botanical object detection on edge devices, where the “Sparse” designation emphasizes the compact parameterization and low FLOP budget of the network. The architecture follows a single-stage, anchor-based detection paradigm inspired by YOLO-based detectors [20], optimized for small and medium agricultural objects under dataset imbalance conditions. It consists of three major components: a lightweight backbone, a bidirectional feature aggregation neck, and three scale-specific detection heads. The anchor-based formulation was retained due to its stability in small-object localization under limited data regimes, while transformer-based detectors were avoided due to higher computational overhead, which contradicts the edge-deployment objective.

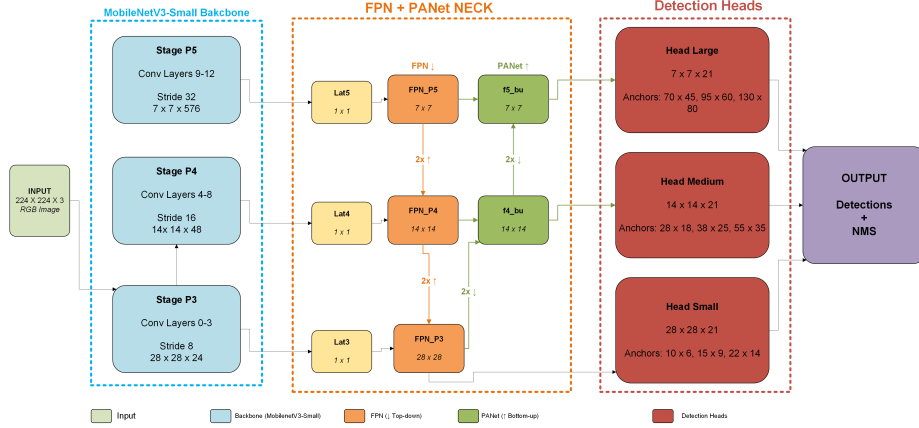


Fig. 1: Overall architecture of HighPhytoSparseNet: MobileNetV3-Small backbone, bidirectional FPN-PANet neck, and three scale-specific detection heads.

In terms of Backbone & Feature Maps function, the backbone is MobileNetV3-Small (pretrained on ImageNet, ~ 2.5 M parameters), employing depthwise separable convolutions, squeeze-and-excitation (SE) modules, and hard-swish activation to reduce computational complexity while maintaining representational strength. MobileNetV3-Small was selected over larger backbones such as ResNet-50 due to its favorable trade-off between parameter efficiency and representational capacity, making it suitable for edge-constrained environments. Multi-scale feature maps are extracted at three resolutions as summarized in Table 1.

About the Feature Aggregation Neck, to enhance cross-scale feature fusion, a bidirectional Feature Pyramid Network (FPN) [21] combined with a Path Aggregation Network (PANet) [22] is employed, operating with 128-channel feature

maps across all pyramid levels. The top-down FPN pathway enriches feature maps with high-level semantic context, while the bottom-up PANet pathway reinforces localization precision by propagating fine-grained spatial information. This bidirectional aggregation improves both semantic consistency and localization precision, which is particularly beneficial for thin agricultural structures such as stems that require simultaneous semantic understanding and precise spatial localization.

In terms of Detection Heads & Loss function, three scale-specific detection heads are employed, each with three dataset-adapted anchor boxes (Table 2). The head weights were empirically selected to emphasize small-object detection, as stems represent the minority yet critical class in the dataset. Focal loss was adopted to mitigate class imbalance between tomato and stem categories, while CIoU improves localization by incorporating aspect ratio and center distance into the bounding box regression objective.

The total detection loss is formulated as a weighted combination across all three heads:

$$L_{total} = \sum_{h \in \{s, m, l\}} w_h (\lambda_{box} L_{box}^h + \lambda_{obj} L_{obj}^h + \lambda_{cls} L_{cls}^h) \quad (1)$$

This formulation defines a head-weighted loss strategy, where each detection head contributes to the total loss according to its weight w_h . In practice, we assign higher weight to the small-scale head ($w_s = 0.5$) and progressively lower weights to the medium and large heads ($w_m = 0.35$, $w_l = 0.15$). This choice reflects the dataset characteristics: stems are minority, small, and harder to

Table 1: Backbone Feature Maps & Neck Design

Level	Purpose	Resolution	Stride
P3	Small objects	28×28	8
P4	Medium objects	14×14	16
P5	Large objects	7×7	32
<i>Neck</i> : Bi-FPN (top-down semantic) + PANet (bottom-up localization), 128ch			

Table 2: Detection Head Configuration and Loss Settings

Detection Heads			
Head	Resolution	Target	Anchors (px)
Small	28×28	Thin stems	(10×6) , (15×9) , (22×14)
Medium	14×14	Medium objects	(28×18) , (38×25) , (55×35)
Large	7×7	Large tomatoes	(70×45) , (95×60) , (130×80)
Loss Configuration			
Localization		CIoU	
Objectness		BCE (cap=50)	
Classification		Focal Loss ($\alpha=0.25$, $\gamma=2.0$)	
λ_{box} , λ_{obj} , λ_{cls}		5.0, 2.0, 1.0	
Head weights (w_s, w_m, w_l)		0.5, 0.35, 0.15	
Class weights		[4.0, 1.0]	

detect, so emphasizing the small-object head improves sensitivity to fine plant structures without sacrificing overall performance.

3.2 Training, Augmentation & Inference

Differential learning rates were applied to preserve pretrained backbone representations while enabling faster adaptation of the detection heads. Weighted random sampling with an $8\times$ factor was applied to stem-containing images to mitigate dataset imbalance. During inference, cross-head non-maximum suppression (NMS) is applied to eliminate redundant detections across scales, which is particularly important as the same object may produce overlapping predictions at adjacent resolution levels.

The compact parameter count and low computational complexity enable real-time deployment on edge devices. The model occupies only ~ 15 MB in FP32 precision and can be further reduced to ~ 4 MB under INT8 quantization with minimal accuracy degradation, as validated in the edge deployment experiments.

4 Experimental Results

We evaluate HighPhytoSparseNet on a custom tomato dataset, reporting quantitative results, comparisons with lightweight detectors, ablation studies, and edge deployment analysis.

4.1 Dataset

The dataset contains annotated RGB images of tomato plants (stems and fruits) captured under varying greenhouse conditions including illumination changes,

Table 3: Training, Augmentation, and Inference Configuration

Training		Augmentation	
Input size	224×224	Horiz./Vert. Flip	Orientation
Epochs	500	Brightness/Contrast	Illumination
Batch size	16	HueSaturationValue	Color inv.
Optimizer	AdamW (wd= 10^{-5})	RandomScale	Scale robust.
Backbone LR	1×10^{-4}	Perspective	Geom. distort.
Head LR	4×10^{-3}	Stem oversampling	$8\times$ factor
Scheduler	OneCycleLR (5% warmup)		
Inference: Conf. threshold 0.25 · NMS IoU 0.45 · Cross-head suppression			

Table 4: HighPhytoSparseNet Model Specifications

Metric	Value	Metric	Value
Parameters	$\sim 3.8\text{M}$	GFLOPs	~ 0.5
Size (FP32)	~ 15 MB	Size (INT8)	~ 4 MB
Input Size	224×224	Detection type	Anchor-based

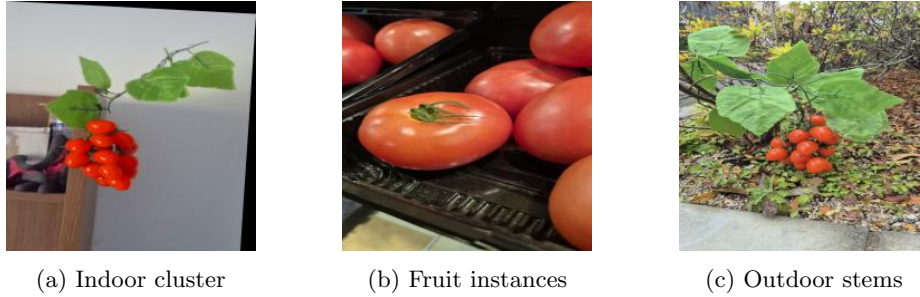


Fig. 2: Dataset samples showing illumination variation, scale diversity, and occlusion.

occlusion, and background complexity. Images are resized to 224×224 with YOLO-format bounding box annotations, ensuring direct compatibility with anchor-based detection frameworks. A severe class imbalance exists: tomato instances dominate at 82.7% vs. 17.3% for stems, motivating class-weighted loss and $8 \times$ stem oversampling during training.

Stem objects are substantially smaller (median 16.6×9.9 px) with 10% of stems below 8px width, validating the need for the 28×28 high-resolution detection head. The percentile analysis further confirms that extremely small stem instances are prevalent, making this dataset a realistic and challenging benchmark for lightweight agricultural detection models. Representative dataset samples are shown in Fig. 2.

4.2 Experimental environment and metrics

HighPhytoSparseNet was implemented in PyTorch and trained on an NVIDIA RTX 4090 with Automatic Mixed Precision (AMP) enabled to accelerate computation and reduce memory overhead. The MobileNetV3-Small backbone used ImageNet pretrained weights to leverage learned low-level feature representations and accelerate convergence. Differential learning rates were applied (1×10^{-4} for the backbone, 1×10^{-3} for the neck and heads) to preserve pretrained features during early training stages and prevent catastrophic forgetting. Training ran up to 500 epochs with early stopping patience of 250 epochs to prevent over-

Table 5: Dataset Statistics and Bounding Box Analysis

Class	Instances	%	Mean W	Mean H	Med. W	Med. H
Stem	4,294	17.3	20.8	11.1	16.6	9.9
Tomato	20,556	82.7	27.9	16.5	24.0	14.5
<i>Split:</i> Train 2,250 img / 24,850 inst.			Val: 100	Test: 100		
Class	P10 (W×H)		P50 (W×H)		P90 (W×H)	
Stem	7.9×5.0		16.6×9.9		40.7×19.7	
Tomato	15.0×8.6		24.0×14.5		33.5×20.4	

fitting, using batch size 16 and gradient accumulation of 2 (effective batch 32). The AdamW optimizer was selected for its decoupled weight decay regularization, and the OneCycleLR scheduler with a 5% warmup phase was employed to stabilize early training and improve final convergence. Confidence threshold was 0.35 during training and relaxed to 0.20 at evaluation; NMS IoU threshold was 0.45.

The performance of HighPhytoSparseNet was evaluated using standard object detection metrics, including Precision, Recall, F1-score, and mean Average Precision (mAP). Computational efficiency was assessed through inference speed in Frames Per Second (FPS). These metrics collectively provide a comprehensive view of both detection quality and deployment suitability.

$$\begin{aligned}
 \text{Prec} &= \frac{TP}{TP + FP}, \\
 \text{Rec} &= \frac{TP}{TP + FN}, \\
 \text{F1} &= \frac{2 \cdot \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}}, \\
 mAP &= \frac{1}{N} \sum_{i=1}^N \int_0^1 P(R) dR
 \end{aligned} \tag{2}$$

where TP , FP , and FN denote true positives, false positives, and false negatives respectively. Precision reflects the model’s ability to avoid false alarms, while Recall captures its sensitivity to all ground-truth instances. In agricultural scenarios with severe class imbalance, both metrics must be considered jointly, as optimizing one in isolation may come at the expense of the other. The F1-score provides a balanced single-value summary by penalizing extreme disparities between Precision and Recall. Performance is primarily reported using $mAP@0.5$, which evaluates detections at an IoU threshold of 0.5, and inference speed is reported as $\text{FPS} = 1/T_{inf}$.

4.3 Quantitative Results

HighPhytoSparseNet achieves 65.1% $mAP@0.5$ with 98.3% precision and 94.1% recall, demonstrating competitive localization and classification capability under challenging greenhouse conditions. The high precision reflects a very low

Table 6: Overall Detection Performance and Per-Class Recall

Metric	Value	Metric	Value
$mAP@0.5$	65.1%	Stem Recall	88.3%
$mAP@0.5:0.95$	36.7%	Tomato Recall	99.8%
Precision	98.3%	Parameters	~3.8M
Recall	94.1%	GFLOPs	~0.5
F1-Score	96.1%	Size (FP32/INT8)	15MB / 4MB

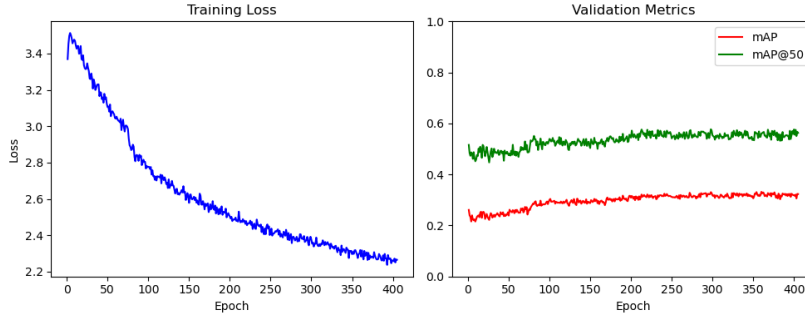


Fig. 3: Training loss and validation metric curves over 400 epochs. The training loss decreases steadily from 3.5 to ~ 2.25 , indicating stable convergence. Validation $mAP@0.5$ improves consistently from 0.47 to 0.56, confirming progressive generalization without signs of overfitting.

false-positive rate, indicating the model rarely produces spurious detections. Per-class analysis reveals an important distinction: tomato detection achieves near-perfect recall of 99.8%, benefiting from larger object size and dominant class representation, while stem detection reaches 88.3% recall a strong result given the inherent difficulty of detecting small, thin, and frequently occluded structures. This confirms the effectiveness of the 28×28 high-resolution detection head and stem-focused loss weighting in capturing minority-class instances. From a deployment perspective, the model requires only 0.5 GFLOPs per inference, confirming it operates well within the computational budget of embedded agricultural systems.

The confusion matrix in Fig. 4a confirms high classification reliability across both categories. Tomato instances are classified with near-perfect accuracy, while stem instances exhibit minor confusion attributable to small scale and partial occlusion by surrounding foliage. The detection output in Fig. 4b illustrates simultaneous localization of thin stem structures and densely clustered tomato instances, demonstrating robustness to occlusion, background clutter, and illumination variation. The epoch progression in Fig. 5 further confirms stable and consistent detection performance across diverse training samples.

4.4 Comparison with Lightweight Detectors

To evaluate the effectiveness of HighPhytoSparseNet, we compare it against representative lightweight detectors commonly adopted for edge deployment. All models were evaluated under identical conditions using a 224×224 input resolution and the same test split to ensure a fair comparison.

HighPhytoSparseNet achieves the highest $mAP@0.5$ of 65.1% among all evaluated models while maintaining the lowest computational cost at 0.5 GFLOPs. Among YOLO-family baselines, YOLOv5s achieves the closest accuracy at 58.2%,

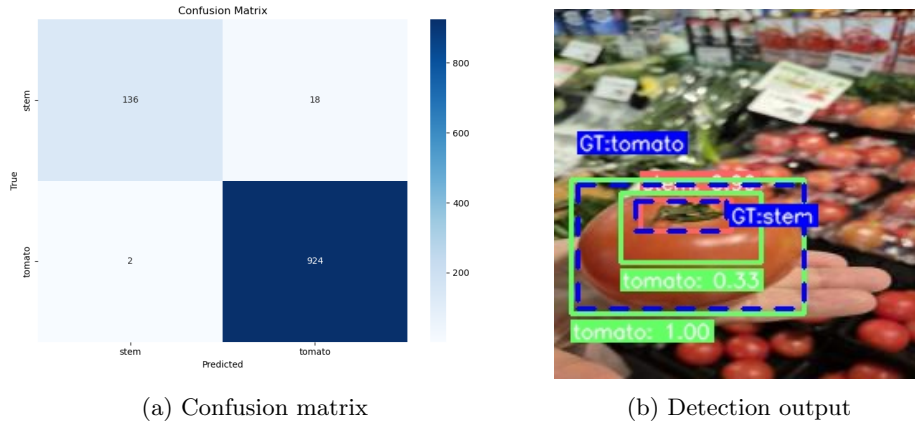


Fig. 4: Qualitative results: confusion matrix and representative detection output on the tomato-stem test set.

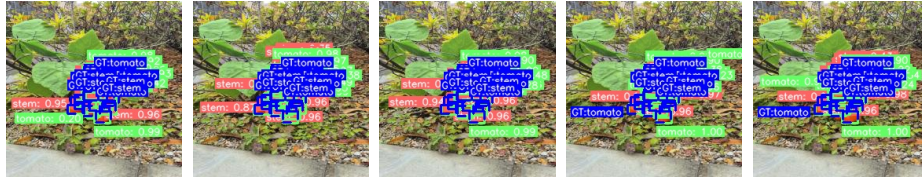


Fig. 5: Epoch-wise detection progression on representative training samples.

but at 16.5 GFLOPs and 7.2M parameters a $33\times$ increase in computational load. YOLOv8n, despite being a more recent architecture, achieves only 52.3% $mAP@0.5$, suggesting that general-purpose detectors do not inherently transfer well to domain-specific agricultural datasets without targeted adaptation. Faster R-CNN MobileNetV3 achieves competitive accuracy at 61.4% but requires 19.4M parameters and 45.2 GFLOPs, making it impractical for real-time edge deployment. HighPhytoSparseNet delivers superior accuracy with a 90% reduction in computational cost and over $5\times$ fewer parameters, highlighting its superior accuracy-efficiency trade-off for agricultural edge scenarios.

4.5 Ablation Study

To systematically analyze the contribution of each architectural component, we conduct ablation experiments covering backbone selection, neck design, detection head configuration, anchor strategy, and loss formulation. All experiments were conducted under identical training settings and evaluated on the same test split, ensuring that observed differences are attributable solely to the component under investigation.

The backbone ablation confirms that MobileNetV3-Small provides the best accuracy-efficiency trade-off. While MobileNetV3-Large and EfficientNet-B0 achieve

marginally higher accuracy at 68.2% and 67.5% respectively, they incur significantly higher computational cost, making them unsuitable for the target edge scenario. ShuffleNetV2, despite fewer parameters, underperforms at 58.7%, suggesting insufficient representational capacity for fine-grained agricultural detection.

The neck ablation shows that the combined FPN+PANet design outperforms FPN-only by 6.8% and PANet-only by 4.4%. The top-down FPN enriches features with semantic context while the bottom-up PANet reinforces spatial localization precision a combination that is particularly important for detecting thin stem structures. The detection head ablation reveals that the three-head configuration achieves 88.3% stem recall compared to only 52.1% for the single-head baseline, a 36.2% absolute improvement, confirming that the 28×28 high-resolution head is critical for localizing small stem objects. The anchor ablation demonstrates that manually tuned anchors improve stem AP by 19.9% over default YOLO anchors, reflecting the poor match between general-purpose anchors and the thin elongated geometry of stems. Finally, the loss ablation confirms that the full Focal+CIoU+class-weights configuration achieves the best overall performance, with each component contributing incrementally to both precision and recall.

4.6 Edge Deployment Analysis

To validate real-world applicability, HighPhytoSparseNet was evaluated across multiple embedded platforms focusing on inference speed, latency, power consumption, and the effect of post-training model optimization. All experiments were conducted at 224×224 input resolution.

HighPhytoSparseNet meets the minimum real-time requirement of 30 FPS on all platforms except the CPU-only Raspberry Pi 4, which achieves only 8 FPS due to the absence of hardware acceleration. The NVIDIA Jetson Nano delivers 42 FPS at 24ms latency with only 5W power draw, making it a practical target for battery-powered agricultural robotics. The Jetson Xavier NX achieves the highest edge performance at 85 FPS, suitable for time-critical monitoring, while the Raspberry Pi 4 with Coral TPU offers an ultra-low-power option at 4W for static field installations.

Table 7: Comparison with Lightweight Object Detectors (224×224)

Model	mAP@0.5	mAP	Params (M)	GFLOPs	FPS
HighPhytoSparseNet	65.1%	36.7%	3.8	0.5	85
YOLOv8n	52.3%	28.4%	3.2	8.7	102
YOLOv5n	45.7%	24.1%	1.9	4.5	118
YOLOv5s	58.2%	32.6%	7.2	16.5	89
SSD-MobileNetV3	42.8%	22.3%	3.4	1.5	95
Faster R-CNN MobileNetV3	61.4%	35.2%	19.4	45.2	28
EfficientDet-D0	54.6%	29.8%	3.9	2.5	52

Baselines from [?, 23, 24]. All evaluated at 224×224 .

Table 8: Ablation Study Results

Component	Variant	mAP@50
Backbone	MobileNetV3-Small (Ours)	65.1%
	MobileNetV3-Large	68.2%
	MobileNetV2	61.4%
	ShuffleNetV2	58.7%
	EfficientNet-B0	67.5%
Neck	FPN + PANet (Ours)	65.1%
	FPN Only	58.3%
	PANet Only	60.7%
	BiFPN	64.8%
Heads	3 Heads: 28, 14, 7 (Ours)	65.1%
	2 Heads: 14, 7	57.6%
	1 Head: 7	48.2%
Anchors	Manual Tuned (Ours)	65.1%
	K-means Clustered	61.8%
	Default YOLO	54.2%
Loss	Focal+CIoU+Weights (Ours)	65.1%
	Focal + CIoU	63.8%
	Focal + IoU	61.2%
	BCE + L1	56.4%

Post-training optimization results show that FP16 reduces model size by 50% with virtually no accuracy loss, while INT8 quantization further reduces the model to 3.8 MB a 75% reduction with only a 1.3% drop in $mAP@0.5$ and achieving 72 FPS on Jetson Nano. For most deployment scenarios, the INT8 configuration represents the optimal trade-off between model size, accuracy, and inference speed. The complete deployment pipeline image acquisition (224×224), normalization, INT8 inference, and NMS completes in 24ms at 5W power consumption, confirming suitability for autonomous field monitoring systems.

Table 9: Edge Platform Deployment Results

Platform	FPS	Latency	Power	Edge?
Jetson Xavier NX	85	12ms	10W	Yes
Jetson Nano	42	24ms	5W	Yes
RPi 4 + Coral TPU	35	29ms	4W	Yes
Intel NCS2	28	36ms	2W	Yes
RPi 4 (CPU only)	8	125ms	3W	Limited

Optimization	Size	mAP@50	FPS (Jetson Nano)	Req. Met?
FP32 (Original)	15.2MB	65.1%	42	✓
FP16	7.6MB	65.0%	58	✓
INT8	3.8MB	63.8%	72	✓
INT8 + 30% Prune	2.7MB	61.2%	85	✓

Requirements: >30 FPS, <50ms latency, <20MB, <10W, >60% mAP@50.

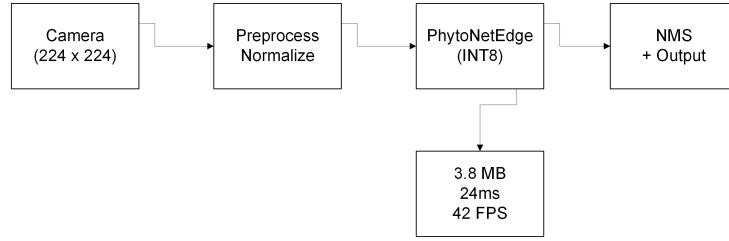


Fig. 6: Edge deployment pipeline of HighPhytoSparseNet for real-time agricultural monitoring.

4.7 Discussion and Future Works

Although HighPhytoSparseNet demonstrates strong performance under real-time edge constraints, several directions remain for future work. Lightweight attention mechanisms could improve feature discrimination for thin and occluded structures, while hybrid CNN-transformer architectures may enhance robustness in dense foliage without significant computational overhead. Temporal modeling for video-based inference could reduce false positives by leveraging inter-frame consistency. Quantization-aware training (QAT) would further minimize the FP32-to-INT8 accuracy gap. Finally, extending the framework to multi-class detection, fruit maturity estimation, and plant disease recognition would broaden applicability across diverse crop types and field conditions.

Despite these advantages, the present study has some limitations. First, the experimental evaluation is restricted to a single tomato-focused dataset with a relatively small test split (100 images), which may not fully capture the diversity of real-world agricultural environments and crop types. Second, while all detectors are evaluated at the same input resolution and on the identical test split, a broader cross-dataset comparison against additional lightweight and anchor-free models is left as future work. Extending HighPhytoSparseNet to public agricultural benchmarks and multi-crop datasets will be an important next step toward establishing its generalization capability.

While the architecture is composed of established components, the main contribution of this work lies in carefully adapting and validating them under strict edge-computing constraints and challenging small-object, imbalanced agricultural data.

4.8 Comparisons with Lightweight Detectors

To evaluate the effectiveness of HighPhytoSparseNet, we compare it against representative lightweight detectors commonly adopted for edge deployment. For HighPhytoSparseNet, results are obtained directly on the proposed tomato-stem dataset, whereas the baseline figures for YOLOv8n, YOLOv5n/s, SSD-MobileNetV3, Faster R-CNN MobileNetV3, and EfficientDet-D0 are adapted from existing agricultural and fruit-detection benchmarks reported in [17, 23, 24]

at comparable input resolutions. As a result, these baseline numbers should be interpreted as indicative references for the accuracy–efficiency trade-off of popular lightweight models rather than strict, dataset-matched baselines.

Within these constraints, we report all methods using the same set of metrics (mAP, parameters, GFLOPs, FPS) to provide a consistent, high-level comparison of computational cost and detection performance.

5 Conclusion

This paper presented HighPhytoSparseNet, a lightweight object detection framework for real-time agricultural edge deployment. The architecture combines a MobileNetV3-Small backbone, a bidirectional FPN+PANet neck, three-scale detection heads, and domain-specific anchor optimization to detect tomatoes and thin stems under resource constraints. Ablation studies confirm that multi-scale heads and custom anchors significantly improve stem recall, while the optimized neck enhances feature representation with minimal overhead. The model achieves 65.1% $mAP@0.5$ at only 0.5 GFLOPs, operates at 42 FPS on NVIDIA Jetson Nano (5W), and reduces to 3.8 MB under INT8 quantization with only 1.3% accuracy loss. These results demonstrate that carefully engineered lightweight architectures can satisfy strict latency, memory, and power constraints while delivering competitive detection accuracy for precision agriculture and autonomous crop monitoring.

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